

ASSIGNMENT II

GGSIU - END TERM QUESTION PAPER 2023-24

Class Teacher: Dr. Monica Gupta

Section: 2nd Year EC3

Paper Code: ECC-214

Subject: Analog Electronics-II

Time: 3 Hours

Maximum Marks: 75

Note: Attempt all 9 questions

Q1 (3×5 = 15)

Answer the following:

- a) Draw the equivalent circuit of an Op-Amp. Explain the concept of virtual ground in Op-Amp.
 - b) Show that an Op-Amp can be used as a switch and as an amplifier.
 - c) What is differential amplifier? Why are differential amplifiers preferred over single-ended amplifiers?
 - d) Explain how do you achieve zero crossing detector using a comparator.
 - e) Discuss the conditions for self-sustained oscillation in feedback amplifiers.
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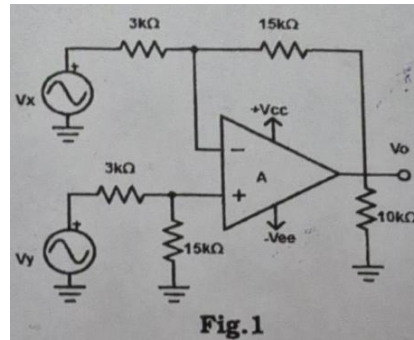
UNIT-I

Q2

- a) What are the characteristics of an ideal Op-Amp? Give typical values for Op-Amp IC 741. (8)
 - b) Draw the block diagram of the operational amplifier and explain the function of each block. (7)
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Q3

- a) Draw the circuit diagram of a closed loop non-inverting amplifier using ideal Op-Amp and derive the expression for the voltage gain. Explain voltage follower circuit with one application. (8)
- b) Find the output voltages V_o of the circuits shown in Fig.1 (Assume that the Op-Amp is ideal) $V_x = 3.9$ Vpp and $V_y = 5.5$ Vpp sine wave at 1 kHz. (7)



UNIT-II

Q4

a) What is slew rate of an Op-Amp? Find an expression for the maximum frequency of the input sine wave f_{max} for undistorted output voltage of peak value V_m .

An amplifier using an Op-Amp with a slew rate $SR = 1 \text{ V}/\mu\text{s}$ has a gain of 40 dB. If this amplifier has to faithfully amplify sinusoidal signals from 10 Hz to 20 kHz, without introducing any slew-rate-induced distortion, calculate the maximum value of the input signal that can be applied without distorting the output. (8)

b) Explain the terms:

- (i) Input bias current
- (ii) Input offset current
- (iii) Input offset voltage
- (iv) Total output offset voltage (7)

Q5

a) Explain how the value of R_{comp} is chosen to nullify the effect of input bias current in a non-ideal Op-Amp. Derive the equation for the value of R_{comp} . (8)

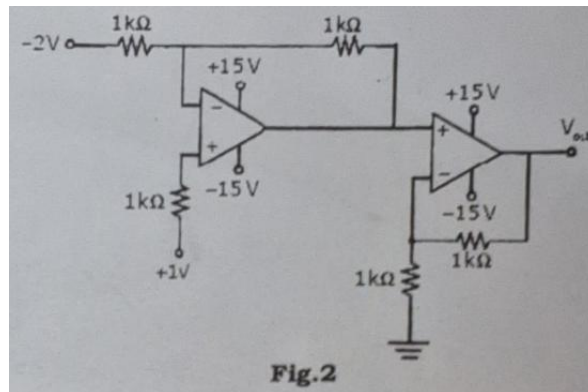
b) Explain the importance of the parameter CMRR of an Op-Amp. An Op-Amp has 100 dB differential gain and its CMRR is 120 dB. Calculate common mode gain. (7)

UNIT-III

Q6

a) Draw the circuit of the Schmitt trigger and explain its working. Why hysteresis is desirable in Schmitt trigger circuits? Derive an expression for hysteresis. (8)

b) Find the output voltages V_o of the circuits shown in Fig.2 (Assume that the Op-Amps are ideal). (7)



Q7

- a) Draw the circuit of a triangular waveform generator using minimum components. Derive an expression for its frequency of oscillation. How this circuit is converted into a ramp generator? (8)
 - b) Draw the circuit of a basic integrator and explain its working. Explain how the gain of an integrator at low frequency can be limited to avoid saturation problem. (7)
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UNIT-IV

Q8

- a) Design a second order low pass filter with upper cut-off frequency 1 kHz. Label the component values in the diagram. (8)
 - b) Explain Butterworth and Chebyshev filter approximation. (7)
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Q9

- a) Explain the working of monostable multivibrator with 555 timer IC. Derive an expression for its output frequency. (8)
- b) Draw the block schematic of PLL and explain the function of each block. (7)